



Chapter 6

Modern Symmetric Block Ciphers: Modes of Operations

Outlines

1. Importance of different modes of operation in block ciphers.
2. Block cipher modes of operation.
3. Introduction to stream ciphers.

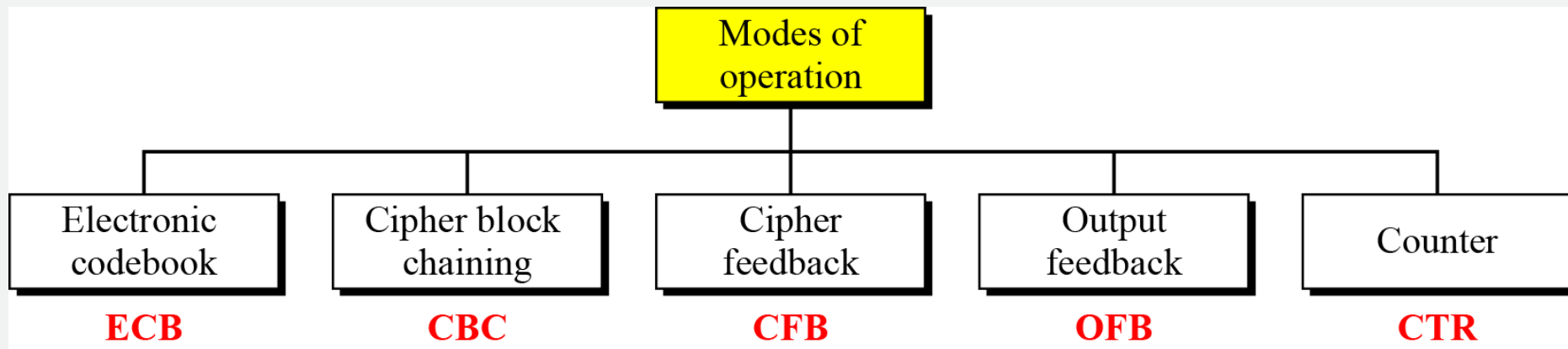
Learning Objectives

In the end of this Chapter 6, students will be able:

1. To understand the **importance** of **different modes of operation** in block ciphers in applications.
2. To learn and perform different modes of operation in block ciphers.
3. To learn the basic concepts and applications of stream ciphers.

Modern Block Cipher Modes of Operations

Motivation & Overview



Modern Block Cipher – Motivation

- ❖ As we learned in Chapter 4 and Chapter 5, block ciphers encrypt data blocks of **fixed sizes**:
 - DES encrypts 64-bit block using 56-bit key.
 - AES encrypts 128-bit block using either 128-bit, 192-bit, or 256-bit key.
- ❖ In practice, **most plaintext blocks do not always exhibit such a bit size**.
- ❖ Hence, there is a need to search for alternatives techniques to encrypt/decrypt arbitrary length of data block.

Modern Block Cipher – Overview

- ❖ **NIST** initially defines **4 modes of operation (in 1981)** to enhance the effect of a cryptographic algorithm, or adapting the algorithm for an application, such as applying block cipher to data block or data stream.
- ❖ The list is then updated to **5 modes** to allow the use of symmetric block cipher including 3DES and AES.
- ❖ The **5 modes** of operation for the purpose of providing confidentiality:
 1. *Electronic Code Book (ECB)*
 2. *Cipher Block Chaining (CBC)*
 3. *Cipher FeedBack (CFB)*
 4. *Output FeedBack (OFB)*
 5. *Counter (CTR)* – **this was added in 2001**

Modes of Operation 1:

Electronic Code Book (ECB)

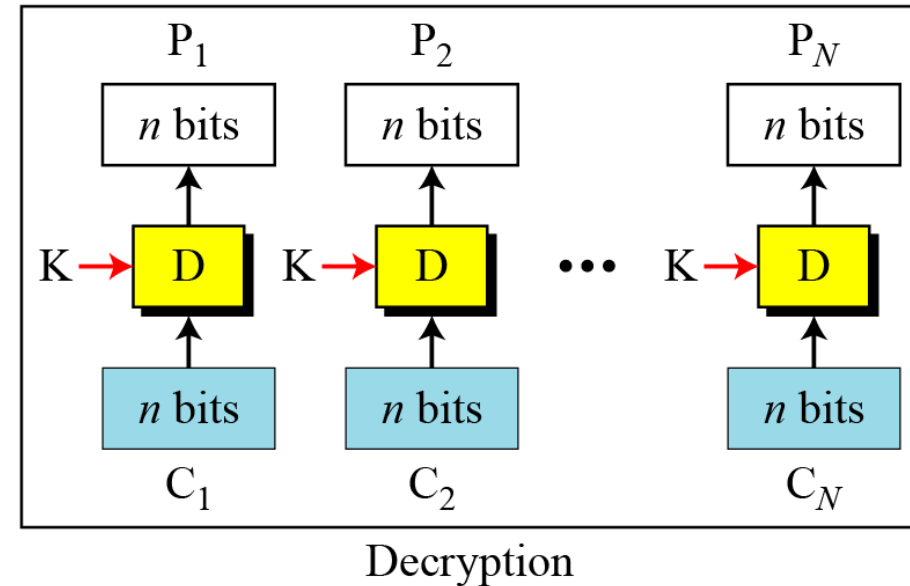
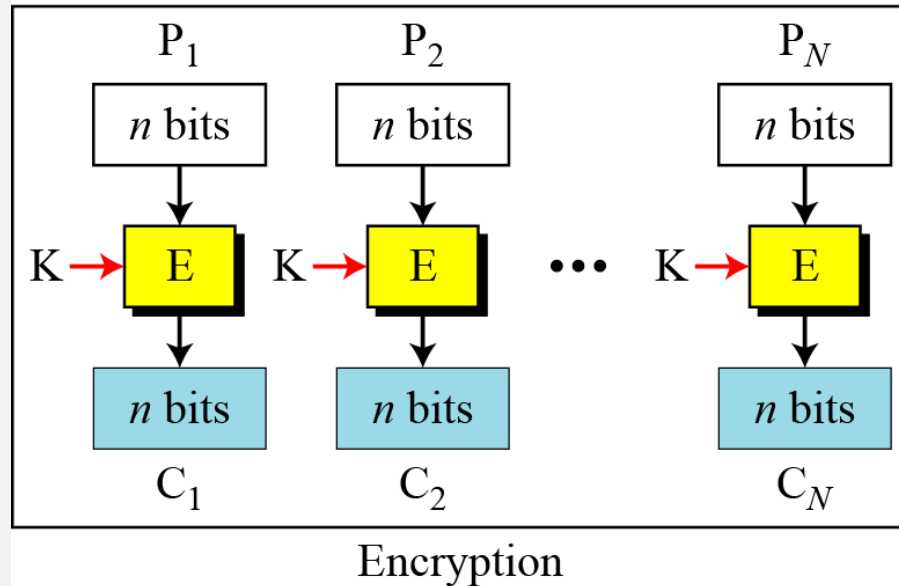
Electronic Code Book (ECB)

- The simplest mode of operation, the plaintext is handled **one block at a time** and is encrypted using the same key.
- It is called '**codebook**' due to unique ciphertext for every b -bit block of plaintext for a given key.
- In other words, one can pre-compile 2^K codebooks, in which each codebook has 2^n entries in two column (each entry with plaintext-ciphertext blocks).
- However, **if K and n are large**, then precompiling and maintaining the codebook are **impractical**.

Electronic Code Book (ECB) – Operation

- **Encryption:** $C_i = E_K(P_i)$
- **Decryption:** $P_i = D_K(C_i)$

E: Encryption D: Decryption
 P_i : Plaintext block i C_i : Ciphertext block i
K: Secret key



Electronic Code Book (ECB) – Ciphertext Stealing

- Once **the size of the block is decide**, let say n -bit, the plaintext is then separated into n -bit long per block.
- If the **LAST block** is not n -bit long, then **padding** (**by inserting 0's**) should be done. However, padding may not always be possible.
- Hence, the technique of **ciphertext stealing (CTS)** is used to resolve this issue. That is, encrypting **without padding** the last block. This technique requires the last 2 plaintext blocks to work.

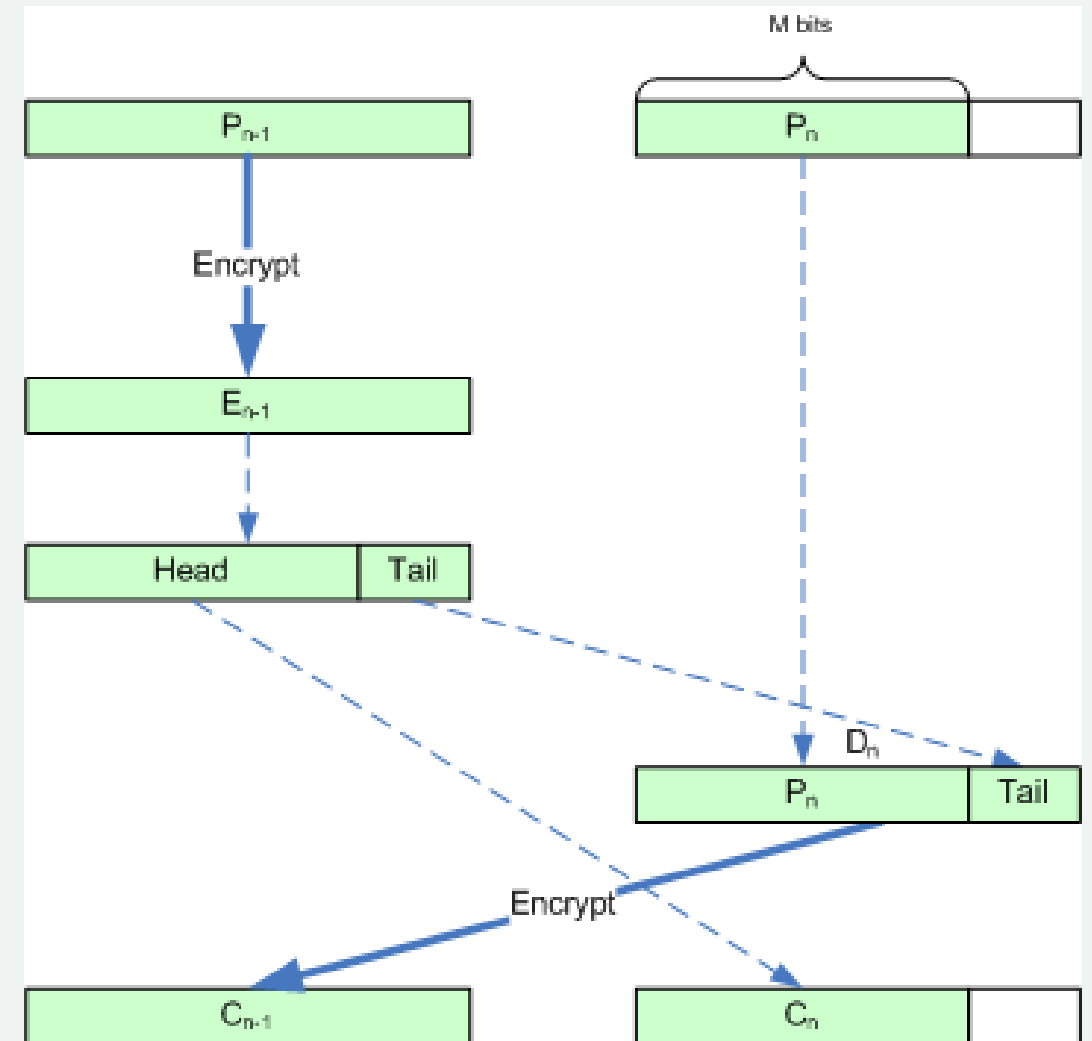
Electronic Code Book (ECB) – Ciphertext Stealing

- The following equations describe the CTS.

$$X = E_K(P_{N-1}) \rightarrow C_N = \text{head}_m(X)$$

$$Y = P_N \parallel \text{tail}_{n-m}(X) \rightarrow C_{N-1} = E_K(Y)$$

- Here, P_N is the **last plaintext block** that lack of some bits, says m -bit long.
- tail** is the **Least Significant Bits (LSB)**, while **head** is the **Most Significant Bits (MSB)**.



Electronic Code Book (ECB) – Security

- Being the simplest mode, ECB suffers from several **security** issues:

1. **Patterns at the block level are preserved.**

- ❖ All **equal blocks** in the plaintext become equal blocks in the ciphertext.
- ❖ For example, if an intruder found that 2nd and 7th ciphertext blocks are the same, then he knows that 2nd and 7th plaintext blocks are the same, too.
- ❖ Sometimes, an intruder can do an **exhaustive search** to decrypt only one of all the blocks to find the important contents of all of them.

Electronic Code Book (ECB) – Security

2. Block Independency issue.

- ❖ An attacker could **exchange/modify** some ciphertext blocks that contains the critical information **without knowing the key**.
- ❖ For example, if the attacker knows that the 10th block in the ciphertext always contain important message (such as amount of money transfer in the transaction), he can then **replace** this block with modified ciphertext block.

Electronic Code Book (ECB) – Applications

- ❑ Secure transmission of **single value**, such as an **encryption key**.
- ❑ **Parallel processing** can be used if we need to create a very huge encrypted database.
- ❑ If the message is short enough to fit in a block, the security issues and propagation errors of ECB are tolerable.
- ❑ ECB is **not recommended** for encryption of **messages of more than one block** to be transferred through an insecure channel.

Modes of Operation 2:

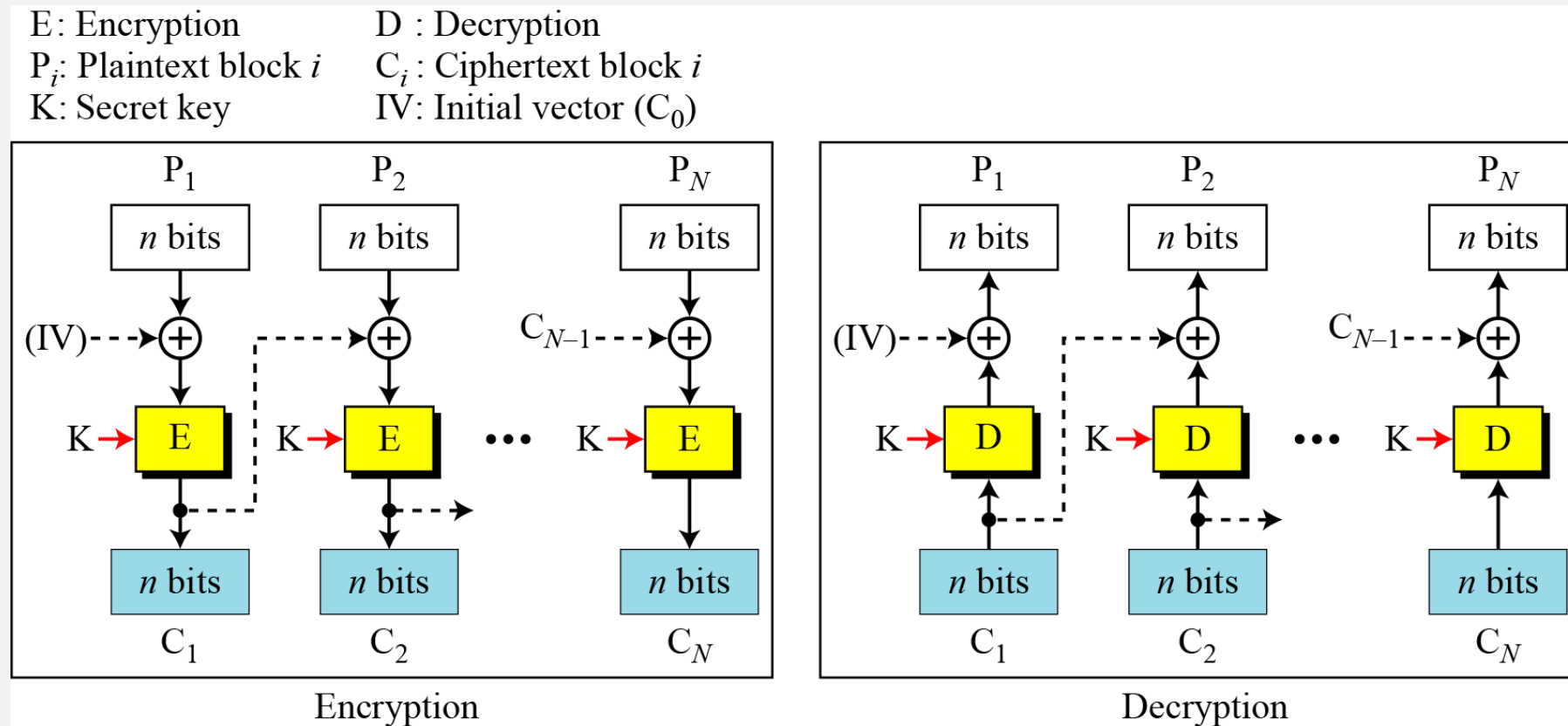
Cipher Block Chaining (CBC)

Cipher Block Chaining (CBC)

- In CBC mode, each plaintext block undergoes **exclusive-OR** operation with the previous ciphertext block before being encrypted.
- When a block is enciphered, the block is sent, and **a copy of it** is kept in memory to be used in the encryption of the next block.
- Since there is no ciphertext block before the first block, a phony block called the **Initialization Vector (IV)** is used, which is **pre-determined (pre-agreed)** between sender and receiver prior to communication.

Cipher Block Chaining (CBC) – Operation

- **Encryption:** $C_i = E_K(P_i \oplus C_{i-1})$ with $C_0 = IV$
- **Decryption:** $P_i = D_K(C_i) \oplus C_{i-1}$ with $C_0 = IV$



Cipher Block Chaining (CBC) – Operation

❖ Initialization Vector (IV):

- A **pseudorandom number** can be selected by the sender and transmitted through a secure channel (for instance, using **ECB mode**)
- It can be part of the **secret key** and should be known by the sender and the receiver.

Cipher Block Chaining (CBC) – Ciphertext Stealing

- Similar to EBC mode, the technique of **ciphertext stealing (CTS)** is also applicable to CBC mode. (Refer to EBC mode for the descriptions)
- The technique is slightly complicated, since CBC mode involves encryption using previous ciphertext block.
- The following equations describe the CTS in CBC mode:

$$\begin{array}{llll} U = P_{N-1} \oplus C_{N-2} & \rightarrow & X = E_K(U) & \rightarrow & C_N = head_m(X). \\ V = P_N \parallel pad_{n-m}(0) & \rightarrow & Y = X \oplus V & \rightarrow & C_{N-1} = E_K(Y) \end{array}$$

Cipher Block Chaining (CBC) – Security

➤ Advantage:

- Since every plaintext block is encrypted using previous ciphertext block in CBC mode, **equal plaintext blocks** belonging to the same message are enciphered into **different ciphertext blocks**. **Patterns** at the block level are **not preserved**.

➤ Security Issues:

1. If two (or first M blocks of different) messages are equal, their encryption is the same if using the **same IV**.
➤ **Recommendation**: Usage of **a timestamp as an IV**.
2. The intruder can add some ciphertext blocks at the end of the ciphertext stream.

Cipher Block Chaining (CBC) – Applications

- ❑ CBC can be used to **encrypt** messages and can be used for **authentication** purposes.
- ❑ However:
 - This technique is **not possible for parallel processing**, since CBC involves chaining mechanism
 - Encryption and decryption for **random-access file records** are **not suitable** too, since they require access to the previous records.

Modes of Operation 3:

Cipher FeedBack (CFB)

Cipher FeedBack (CFB)

- This technique is used when we need to use **DES or AES as secure ciphers**, but the plaintext or ciphertext **block sizes are smaller**.
- For example, to encrypt and decrypt ASCII 8-bit characters.
- In other words, the plaintext or ciphertext blocks with size **r -bit** is encrypted or decrypted using DES or AES key of size **n -bit**, where **$r \leq n$** .

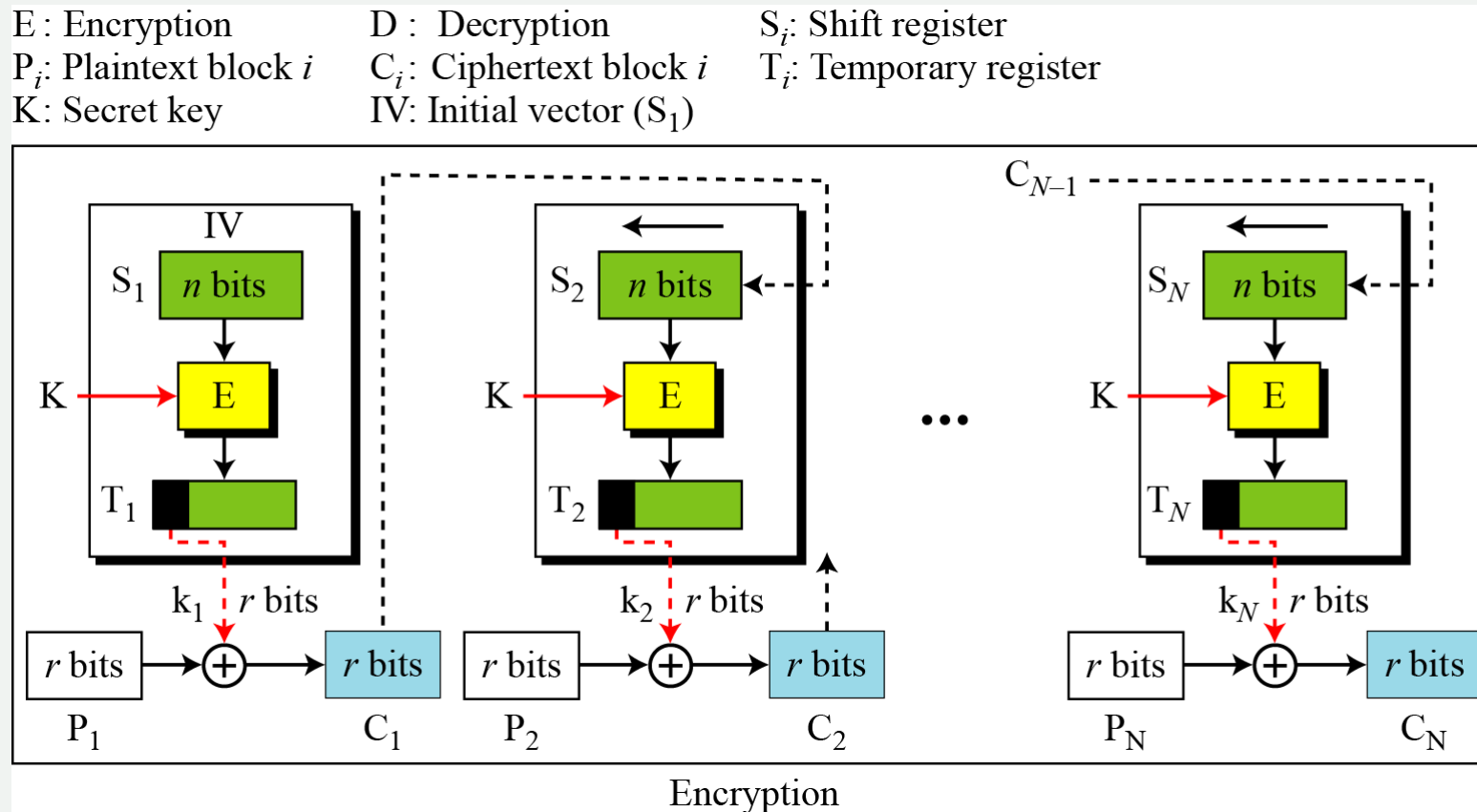
Cipher FeedBack (CFB) – Operation

❖ Idea:

- Use DES or AES to encrypt or decrypt the contents of shift register S of size n -bit.
- **Encryption:** r -bit plaintext block $\oplus$ r -bit of the shift register S . For each block:
 - Shift register S_i is made by shifting the shift register S_{i-1} r -bit to the **LEFT** and filling the rightmost r -bit with C_{i-1} . S_i is then encrypted to T_i .
 - The leftmost r -bit of T_i are $\oplus$ with the plaintext block P_i to produce C_i .
 - S_1 which is not shifted is set to **Initialization Vector (IV)** for the first block.
- **Decryption:** r -bit ciphertext block $\oplus$ r -bit of the shift register S .
 - Roles of plaintext blocks (P_i) and ciphertext blocks (C_i) are **switched**.

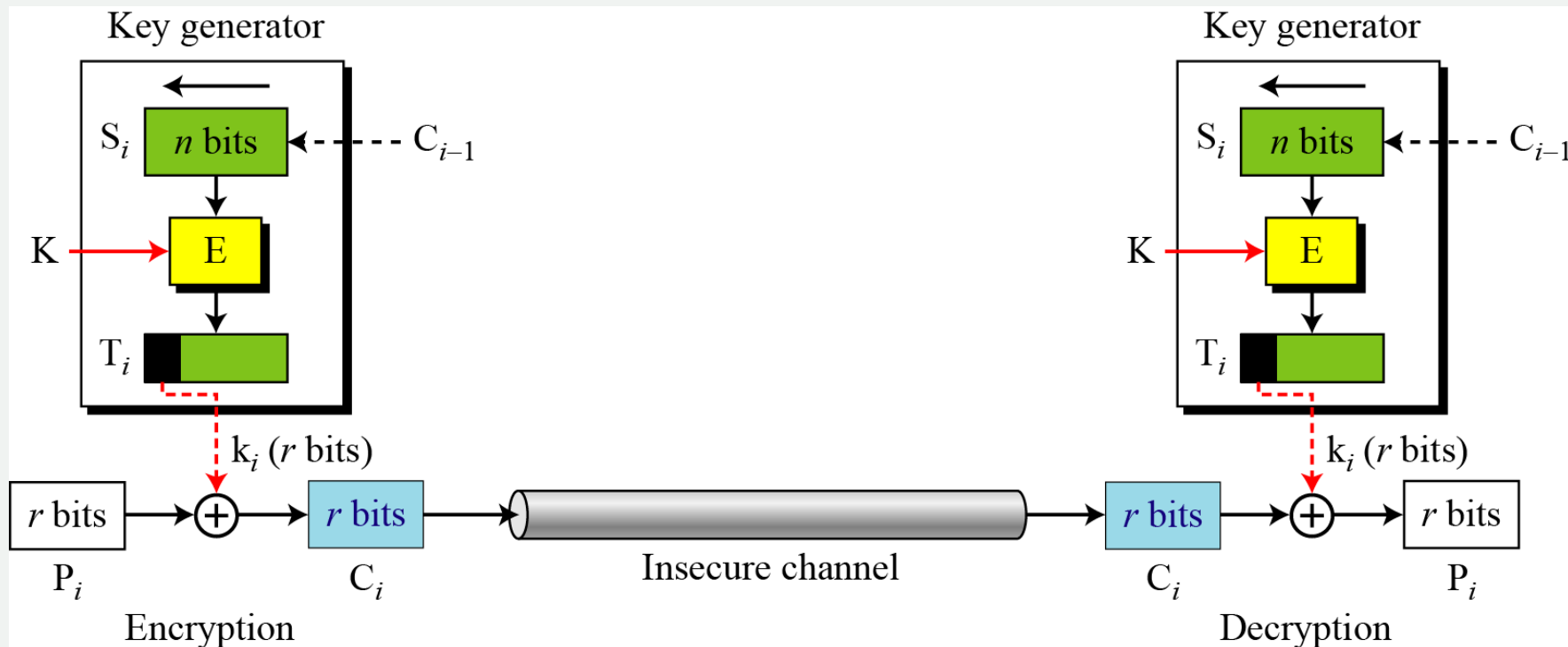
Cipher FeedBack (CFB) – Operation

- **Encryption:** $C_i = P_i \oplus \text{SelectLeft}_r\{\mathbf{E}_K[\text{Shiftleft}_r(S_{i-1})|C_{i-1}]\}$
- **Decryption:** $P_i = C_i \oplus \text{SelectLeft}_r\{\mathbf{E}_K[\text{Shiftleft}_r(S_{i-1})|C_{i-1}]\}$



Cipher FeedBack (CFB) – Stream Cipher

- Although **CFB mode operates using block ciphers** (DES and AES), **the result is a stream cipher**.
- It is in fact a **nonsynchronous stream cipher**, the key stream is dependent on the ciphertext.



Cipher FeedBack (CFB) – Security

❖ Security issues/concerns:

1. Like CBC mode, the patterns at the block level are not preserved.
2. More than one plaintext can be encrypted using the same key, but the value of initialization vector (IV) should be different for each plaintext.
3. An attacker can add some ciphertext block to the end of the ciphertext stream.

Cipher FeedBack (CFB) – Application

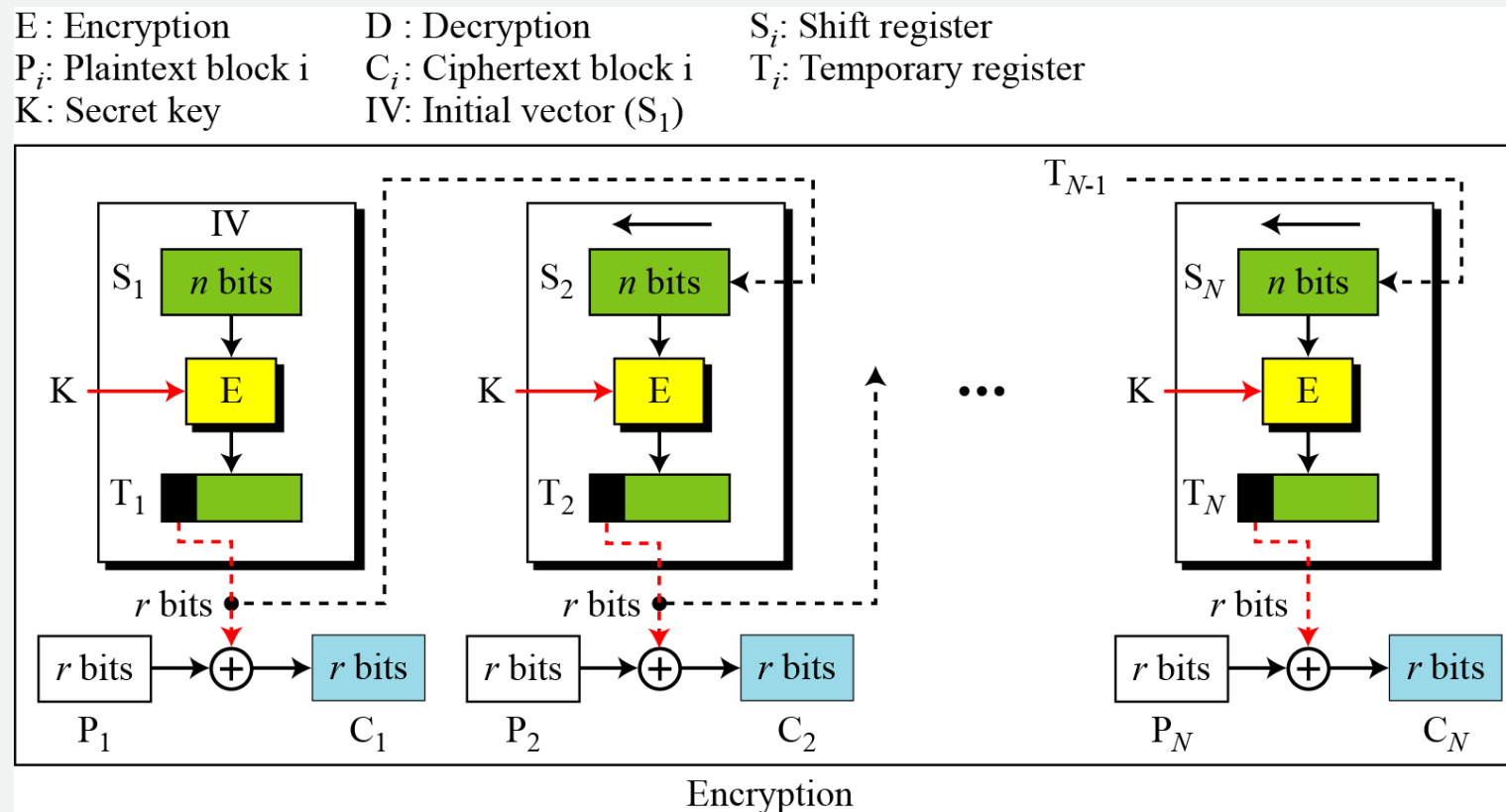
- CFB mode can be used to **encrypt small-size blocks**, such as one character or one bit at a time.
- **Advantages:**
 - ✓ **No padding is needed** since the size of the plaintext block is normally fixed and chosen to fit the data unit to be encrypted.
 - ✓ **No waiting is required** for the system to receive a large data block (64 or 128 bits) to start the encryption.
- **Disadvantage:**
 - ✓ **Less efficient** since the encryption function must be applied for each small block of size r -bit.

Modes of Operation 4:

Output FeedBack (OFB)

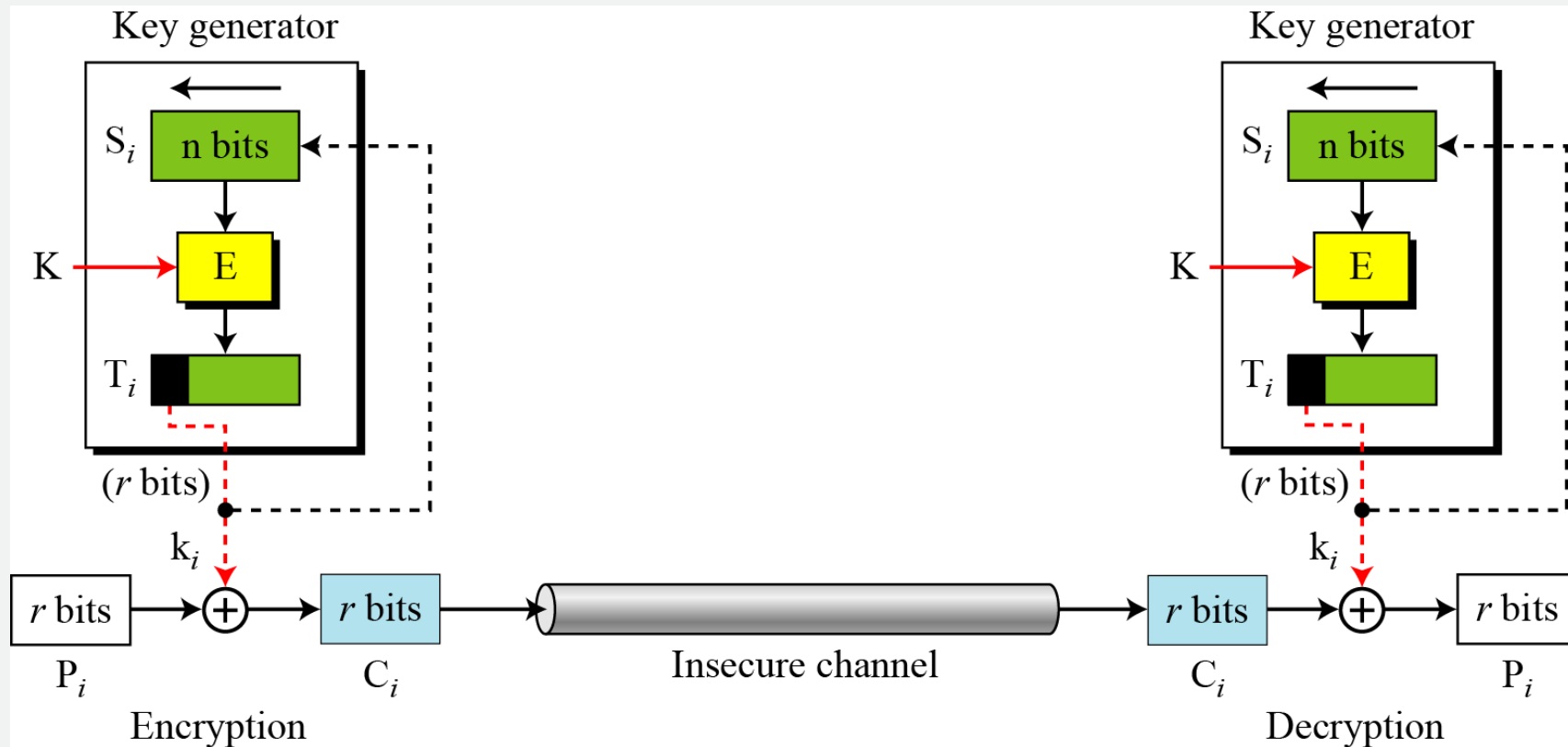
Output FeedBack (OFB) – Operation

- Similar technique to CFB mode, with the difference that each bit in the ciphertext is **independent** of the previous bit or bits. This **avoids error propagation**.



Output FeedBack (OFB) – Stream Cipher

- Similar to CFB mode, the output of the OFB mode is a stream cipher.
- The key stream is **independent** from the plaintext and ciphertext, hence the **stream cipher is synchronous**.



Output FeedBack (OFB) – Security

❖ Security issues/concerns:

1. Similar to CFB mode, the **patterns** at the block level are **not preserved**.
2. The change in the ciphertext affects the plaintext encrypted at the receiver side.

❖ Advantage:

- ✓ A single error in the ciphertext affects only the corresponding bit in the plaintext.

Modes of Operation 5:

Counter (CTR)

Counter (CTR)

- ❖ In this technique, no feedback is involved, and the **pseudorandomness** in the key stream is achieved using a **counter**.
- ❖ An n -bit counter is **initialized to a pre-determined value (IV)** and incremented based on a predefined rule (mod 2^n).
- ❖ The plaintext and ciphertext have the **same block size** as the underlying cipher (such as DES and AES). That is, the n -bit plaintext block is encrypted using n -bit key to produce n -bit ciphertext block.

Counter (CTR) – Operation

- **Encryption:** $C_i = P_i \oplus E_{k_i}(\text{Counter})$
- **Decryption:** $P_i = C_i \oplus E_{k_i}(\text{Counter})$

E : Encryption

P_i : Plaintext block i

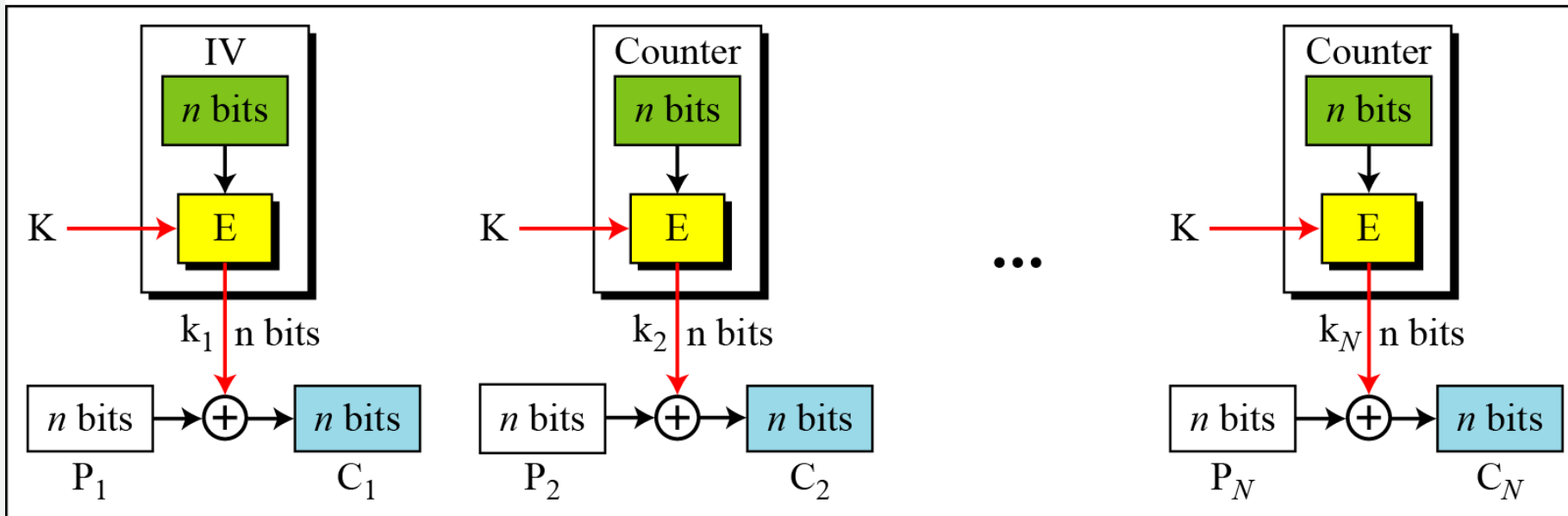
K : Secret key

IV: Initialization vector

C_i : Ciphertext block i

k_i : Encryption key i

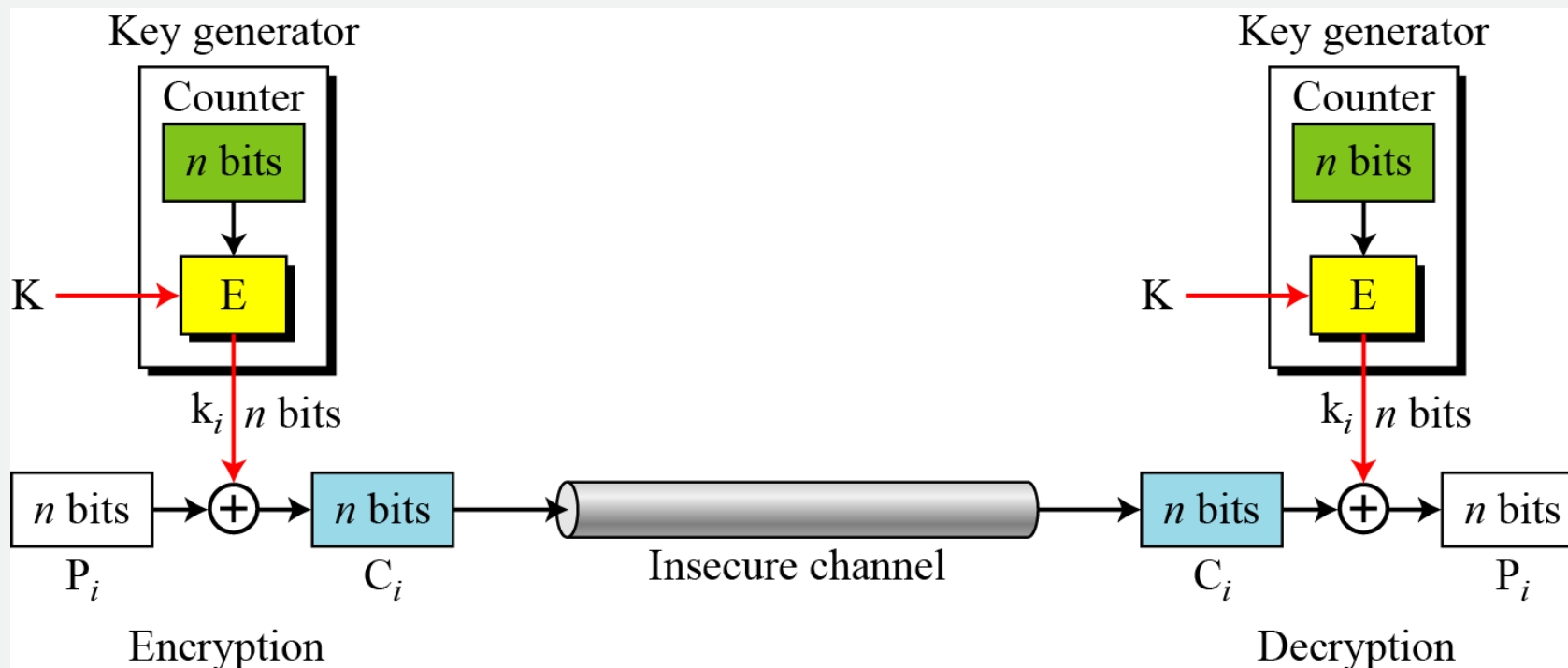
The counter is incremented for each block.



Encryption

Counter (CTR) – Stream Cipher

- CTR mode is actually a stream cipher, as different blocks are $\oplus$ with different keys).



Counter (CTR) – Security

❖ Security issues/concerns:

1. Similar to CFB mode, the **patterns** at the block level are **not preserved**.
2. The change in the ciphertext affects the plaintext encrypted at the receiver side.

❖ Advantage:

- ✓ A single error in the ciphertext affects only the corresponding bit in the plaintext.

Modern Block Cipher

Summary

Modern Block Cipher – Summary

<i>Operation Mode</i>	<i>Description</i>	<i>Type of Result</i>	<i>Data Unit Size</i>
ECB	Each n -bit block is encrypted independently with the same cipher key.	Block cipher	n
CBC	Same as ECB, but each block is first exclusive-ored with the previous ciphertext.	Block cipher	n
CFB	Each r -bit block is exclusive-ored with an r -bit key, which is part of previous cipher text	Stream cipher	$r \leq n$
OFB	Same as CFB, but the shift register is updated by the previous r -bit key.	Stream cipher	$r \leq n$
CTR	Same as OFB, but a counter is used instead of a shift register.	Stream cipher	n

Stream Ciphers

A Brief Introduction

Stream Ciphers – A Brief Introduction

- Although the five modes of operations enable the use of block ciphers for encrypting data in large units and small units, sometimes **pure stream** are needed for encrypting **small units** of data such as characters or bits.
- Hence, **stream ciphers** are more **efficient** for real-time processing.
- Some of the popular and practical stream ciphers:
 - 1. RC4 (Rivest Cipher)** – Widely used in Transport Layer Socket (TLS) and Secure Socket Layer (SSL).
 - 2. A5/1** – Used in Global System for Mobile Communication (GSM).
 - 3. Software Optimized Encryption Algorithm (SEAL)**
 - 4. Salsa20**
 - 5. Very Efficient Substitution Transposition (VEST)**

Main References for Chapter 6

1. Forouzan, B.A. "***Introduction to Cryptography and Network Security***", Third Edition, McGraw-Hill, 2008.
2. Stallings, W. "***Cryptography and Network Security: Principles and Practices***", Fifth Edition, Pearson Prentice Hall, 2011.

~ THE END ☺ ~